



CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– Hanoi, Sep 2025 –

Table of Contents

I. Record of Changes	3
II. Project Management Plan	4
1. Overview	4
1.1 Cost & Time Estimations	4
1.2 Project Objectives	10
1.3 Project Risks	11
2. Management Approach	12
2.1 Project Processes	12
2.2 Quality Management	13
2.3 Project Training Plan	13
3. Responsibility Assignments.....	14
4. Project Communications.....	16
5. Configuration Management.....	16
5.1 Document Management	16
5.2 Source Code Management.....	16
5.3 Tools & Infrastructures	16

I. Record of Changes

Date	A* M, D	In charge	Change Description
10/9/2025	A	QuatBT	Add Project Management Plan
15/9/2025	M	Team G44	Update Project Management Plan
27/9/2025	M	Team G44	Update Project Management Plan
3/10/2025	M	Team G44	Update Project Management Plan
16/10/2025	M	Team G44	Update Project Management Plan
22/10/2025	M	Team G44	Update Project Management Plan
2/11/2025	M	Team G44	Update Project Management Plan
25/11/2025	M	Team G44	Update Project Management Plan
7/12/2025	M	Team G44	Final review and formatting.

*A - Added M - Modified D - Deleted

Definitions and Acronyms

Acronym	Meaning
A	Added
M	Modified
D	Deleted
pds	Person-days
CRUD	Create, Read, Update, Delete
UI	User Interface
APIs	Application Programming Interfaces
QR system	Quick Response system
Genemi AI	Genemi Artificial Intelligence

II. Project Management Plan

1. Overview

1.1 Cost & Time Estimations

#	Work Package	Est. Effort (pds)	Deadline
1	<i>Feature 1: Common Feature</i>	13	
1.1	Login	2	Week 1
1.2	Logout	1	Week 1
1.3	Change Password	2	Week 1
1.4	Forgot Password	2	Week 1
1.5	Register	2	Week 1
1.6	Homepage	2	Week 1
1.7	Search engine course at homepage	2	Week 1
2.	<i>Feature 2: Profile Feature</i>	4	
2.1	View profile	2	Week 1
2.2	Update profile	2	Week 2
3	<i>Feature 3: News Feature</i>	4	
3.1	Create news	1	Week 2
3.2	View news	1	Week 2
3.3	Update news	1	Week 2
3.4	Delete news	1	Week 2
4	<i>Feature 4: Ticket Feature</i>	6	
4.1	Create ticket	1	Week 2
4.2	View detail ticket	1	Week 2
4.3	Update ticket	1	Week 2
4.4	View list ticket	1	Week 2
4.5	Resolve ticket	2	Week 2
5	<i>Feature 5: Forum Feature</i>	15	
5.1	View my post list	2	Week 3

5.2	View post list user	2	Week 3
5.3	Create post	1	Week 3
5.4	Update post	2	Week 3
5.5	Delete post	1	Week 3
5.6	Comment post	1	Week 3
5.7	Resolve comment	1	Week 3
5.8	Like comment	1	Week 3
5.9	Report comment	1	Week 3
5.10	View post list all	1	Week 3
5.11	Manage Forum	2	Week 3
6	Feature 6: Category Feature	7	
6.1	Create category	2	Week 4
6.2	Update category	2	Week 4
6.3	Delete category	1	Week 4
6.4	View category	2	Week 4
7	Feature 7: Forum Category Feature	4	
7.1	Create forum category	1	Week 4
7.2	Update forum category	1	Week 4
7.3	Delete forum category	1	Week 4
7.4	View forum category	1	Week 4
8	Feature 8: SubCategory Feature	4	
8.1	Create subcategory	1	Week 4
8.2	Update subcategory	1	Week 4
8.3	Delete subcategory	1	Week 4
8.4	View subcategory	1	Week 4
9	Feature 9: Notification Feature	5	
9.1	View notify userId	2	Week 5
9.2	View notify detail	1	Week 5
9.3	Send notify system	2	Week 5

10	Feature 10: Course Feature	25	
10.1	View the list of all courses by subcategory	2	Week 5
10.2	View detail course	4	Week 5
10.3	View the list of courses enrolled in by students	3	Week 5
10.4	View the list of all courses for the Moderator	1	Week 5
10.5	View my course list for the Lecturer	1	Week 5
10.6	Create course	3	Week 5
10.7	Update course	3	Week 5
10.8	Delete course	2	Week 5
10.9	Ban/unban course	2	Week 6
10.10	Report course	1	Week 6
10.11	Approve/Reject the course	2	Week 12
10.12	Publish course	1	Week 12
11	Feature 11: Voucher Feature	7	
11.1	View voucher	2	Week 6
11.2	Create voucher	2	Week 6
11.3	Update voucher	2	Week 6
11.4	Disable voucher	1	Week 6
12	Feature 12: Lesson Feature	35	
12.1	View Lesson	3	Week 6
12.2	Create Lesson	4	Week 7
12.3	Update Lesson	4	Week 7
12.4	Delete Lesson	2	Week 7
12.5	Create comment lesson	2	Week 10
12.6	Update comment lesson	2	Week 10
12.7	Delete comment lesson	2	Week 10
12.8	View comment lesson	2	Week 11
12.9	Like comment lesson	2	Week 11
12.10	Resolve comment lesson	2	Week 11

12.11	Lesson-based AI Chat	10	Week 7
13	Feature 13: Cart Feature	6	
13.1	View Cart	1	Week 7
13.2	Add to cart	2	Week 7
13.3	Remove item from cart	1	Week 7
13.4	Enter and apply voucher	2	Week 8
14	Feature 14: Question Bank Feature	15	
14.1	Create a question in the question bank	3	Week 8
14.2	View question bank	1	Week 8
14.3	Update question bank	4	Week 8
14.4	Delete the question from the question bank	2	Week 8
14.5	Add multiple questions from Excel to the question bank	5	Week 8
15	Feature 15: Quiz Feature	27	
15.1	Create quiz	3	Week 8
15.2	View quiz	2	Week 8
15.3	Update quiz	4	Week 8
15.4	Delete quiz	2	Week 8
15.5	Student do quiz	6	Week 12
15.6	View result detail quiz	2	Week 12
15.7	Add question to quiz	4	Week 9
15.8	Update question in quiz	4	Week 9
16	Feature 16: Section Feature	4	
16.1	Create section	1	Week 9
16.2	View section	1	Week 9
16.3	Update section	1	Week 9
16.4	Delete section	1	Week 9
17	Feature 17: Lecturer Application Feature	7	
17.1	Apply lecturer Application	3	Week1

17.2	Manage lecturer application	2	Week1
17.3	View lecturer application	2	Week1
18	Feature 18: Question Feature	20	
18.1	Create question	6	Week1
18.2	View question	4	Week1
18.3	Update question	6	Week1
18.4	Delete question	4	Week1
19	Feature 19: Answer Feature	7	
19.1	Create answer	2	Week1
19.2	View answer	1	Week1
19.3	Update answer	2	Week1
19.4	Delete answer	2	Week1
20	Feature 20: Rating Feature	2	
20.1	Create rating	1	Week1
20.2	View rating	1	Week1
21	Feature 21: Student Progress Feature	3	
21.1	View learning progress	2	Week 10
21.2	Resume Learning	1	Week 10
22	Feature 22: Payment Feature	4	
22.1	Buy course now	2	Week 13
22.2	Buy course in cart	2	Week 13
23	Feature 23: Transaction Feature	6	
23.1	View transaction	2	Week 11
23.2	Remove transaction	2	Week 11
23.3	Create transaction	2	Week 11
24	Feature 24: Banner Feature	9	
24.1	Create banner	2	Week 13
24.2	View banner	2	Week 13
24.3	Update banner	2	Week 13

24.4	Hidden banner	2	Week 13
24.5	Visible banner	1	Week 13
25	Feature 25: User Feature	2	
25.1	User management	2	Week 9
26	Feature 26: Lecturer Feature	4	
26.1	View lecturer	2	Week 10
26.2	Update lecturer	2	Week 10
27	Feature 27: Admin Feature	4	
27.1	Moderator management	2	Week 9
27.2	Lecturer Payroll Report	2	Week 11

#	Work Package	Est. Effort (pds)	Deadline
1	Report 1: Project Introduction	3	Week 14
2	Report 2: Project Management Plan	3	Week 14
3	Report 3: Software Requirement Specification	6	Week 14
4	Report 4: Software Design Document	6	Week 14
5	Report 5: Test Documentation	3	Week 14
6	Report 5.1: Unit Test	6	Week 14
7	Report 5.2: Integration Test	5	Week 14
8	Report 5.3: System Test	5	Week 14
9	Report 6: Software User Guides	3	Week 14
10	Report 7: Final Project Report	5	Week 14
11	Meeting & review	30	Week 14
12	Project Management Overhead	30	Week 14

Total Estimated Effort (person-days)

354 person-day

1.2 Project Objectives

#	Metric	Unit	Planned	Actual	Notes / References
1	Effort Usage	Person-day	350	354	Need to analyze more for the actual effort distribution (as tracked in the Project Tracking, sheet Scope)
2	Review Defects	No of defects	40	55	Actually found more defects (55/40), indicating a strong and effective review process.
3	Unit Test	No of defects	100	80	312 test cases (test functions) successfully passed. The target was to find 100 bugs during unit testing, 80 have been fixed, evidence is described in the following file:
4	Integration Test	No of defects	25	18	130 test cases successfully passed. Low integration defects (18/25) combined with high test case pass rate.
5	System Test	No of defects	15	10	145 test cases successfully passed. System defects are low (10/15) compared to the plan.
6	Acceptance Test	No of defects	5	2	Very low defects found by the end-user (2/5), confirming high output quality.

1.3 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	Lack of specific customers	The system may not meet real learning or teaching needs, reducing usability and adoption	Medium	Conduct early surveys with students, lecturers, and moderators; validate requirements through prototypes and sprint reviews
2	Requirement changes during development	Delays schedule, increases workload, causes rework	Medium	Freeze requirements per increment, apply change-control review, and document all updates clearly
3	Limit team availability due to school schedules	Tasks may not be completed on time, affecting sprint commitments	High	Plan workload based on real weekly capacity, prioritize essential features, and redistribute tasks early
4	Poor communication between team members	Misunderstandings in feature behavior, delays in integration, lower quality	Medium	Hold meetings every evening, provide daily reports, and maintain clear documentation
5	Dependency on third-party services(payment APIs, email sender, storage, QR system, Genemi AI)	Payment or notification features may fail if third parties have downtime	Medium	Choose reliable providers; implement fallback messages; retry mechanisms and error logging
6	Performance issues due to large course data, many users, or heavy queries	Slow loading pages, affecting learning experience and user satisfaction	Low	Optimize database queries; load test high-traffic features (Course, Forum)

2. Management Approach

2.1 Project Processes

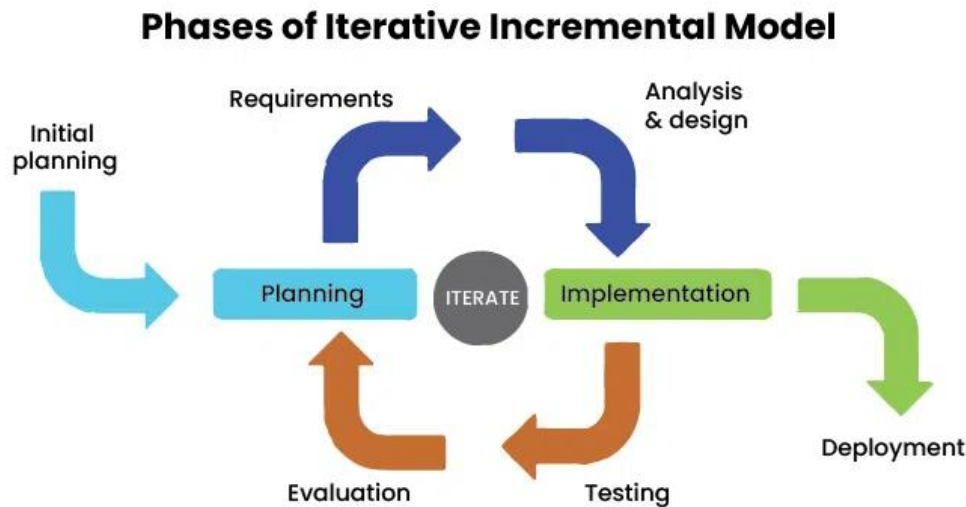


Figure 1 - The process of developing a software project

After evaluating user roles, customer expectations, team availability, and the modular nature of the system requirements, the team selected the Iterative–Incremental model as the most suitable methodology. The system serves students, lecturers, and moderators, all requiring a stable, intuitive platform that evolves continuously. With customers expecting updates every 2–3 weeks and the team having varied schedules, the Iterative–Incremental model ensures flexibility and predictable delivery.

The project is divided into three increments across the 14-week timeline. Increment 1 (Weeks 1–4) delivers the foundational system, including authentication, homepage and shared UI, user profile, news module, ticket management, basic forum interactions, and category structures, ensuring an early functional version for feedback. Increment 2 (Weeks 5–9) expands the system into a complete learning ecosystem by implementing notifications, course modules with CRUD and role-specific behaviors, voucher and cart system, lesson and section management, quiz and question bank, and user/moderator management. Increment 3 (Weeks 10–13) provides all advanced and commercial features, including learning progress tracking, direct and cart-based course purchasing flows, reporting for courses and lessons, quiz-taking and result review, lesson interactions (comment, reply, like, report, resolve), transaction history, administrative functions (salary, feature toggles, course approval/publishing), search engine, drag-and-drop course structure, lecturer dashboard, lecturer profiles, and enrolled-student views.

Week 14 is dedicated to system testing, integration validation, bug fixing, performance checks, and deployment preparation. By delivering fully testable modules at each increment, the team ensures rapid feedback, early defect detection, and steady progress toward project goals. This Iterative–Incremental approach minimizes risk, maximizes efficiency, and fits the constraints and capabilities of the development team while ensuring all objectives are achieved within the 14-week schedule.

2.2 Quality Management

To ensure the quality of the Automated Essay Scoring System and reach the project quality objectives, the team will apply a comprehensive approach including three main stages: Defect Prevention, Reviewing, and Testing.

2.2.1. Defect Prevention (Coding Standards)

Approach: All team members must strictly follow the defined Coding Conventions (naming rules, comment styles, folder structure) for both Frontend and Backend.

Goal: To maintain code consistency and minimize syntax or logic errors during the development phase.

2.2.2. Reviewing (Peer Reviews)

Document Review: Requirements and design documents (SRS, SDS) are reviewed by the Team Leader and Instructor before implementation to ensure clarity.

Code Review: Before merging any code into the main branch (e.g., on GitHub/GitLab), a Pull Request (PR) must be created and reviewed by at least one other team member. This helps detect hidden bugs and optimize code logic.

2.2.3. Testing Strategy The project conducts testing using 3 levels:

Unit Testing: Performed by Developers during coding. We verify individual functions/methods to ensure they work correctly. (Tools: JUnit, NUnit, or Jest).

Integration Testing: Performed by Developers when combining modules (e.g., connecting the Essay Scoring Algorithm with the Database).

System Testing: Performed by the Tester. The entire application is tested as a complete system to verify functionality against the SRS requirements.

2.2.4. Defect Tracking

All defects found are logged into a tracking sheet (e.g., Excel) with priority levels (Low, Medium, High, Critical).

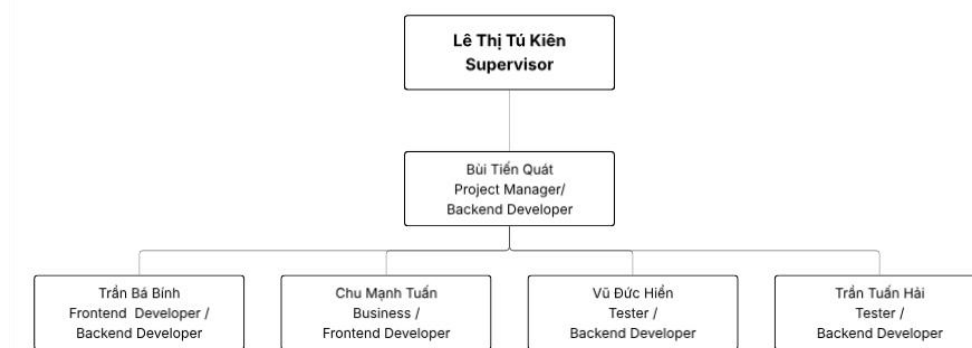
Process: Log Bug -> Assign to Dev -> Fix -> Retest -> Close.

2.3 Project Training Plan

Training Area	Participants	When, Duration
Git & GitHub Workflow(Clone, Branch, Commit, Pull Request)	All team members	Week 1(1 day workshop)
Backend Core(C#, .NET, RESTful API structure)	BinhTB,QuatBT, HienVD, HaiTT	Week 1 - 2(1 week)
Database & ORM(SQL Server, Entity Framework Core)	BinhTB,QuatBT, HienVD, HaiTT	Week 2(1 week)
System Integration(Connecting React with .NET API)	BinhTB,QuatBT, HienVD, HaiTT,TuanCM	Week 3(3 days)
Frontend Development(ReactJS, Hooks, Tailwind CSS)	BinhTB,QuatBT, HaiTT,TuanCM	Week 1 - 2(2 weeks)

Software Testing(Unit Test techniques, Bug logging)	All team members	Week 4(1 week)
---	------------------	----------------

3. Responsibility Assignments



RACI Chart: R~Responsible, A~Accountable, C~Consulted, I~Informed

Work Package	H E 1 7 6 3 8 2	H E 1 7 3 2 4 4	H E 1 8 6 6 2 1	H E 1 7 2 2 3 8	H E 1 6 3 3 4 4
Feature 1: Common Feature	R, A	R	I	C	I
Feature 2: Profile Feature	A	R	I	C	I
Feature 3: News Feature	A	I	R	R, C	I
Feature 4: Ticket Feature	R	I	R	C	I
Feature 5: Forum Feature	A	I	I	R	R
Feature 6: Category Feature	A	I	R	C	R
Feature 7: Forum Category Feature	A	I	R	C	R
Feature 8: SubCategory Feature	A	I	R	C	R
Feature 9: Notification Feature	A	I	R	C	R
Feature 10: Course Feature	R,A	R	I	R, C	I
Feature 11: Voucher Feature	R,A	I	R	C	I
Feature 12: Lesson Feature	R,A	I	I	R, C	I
Feature 13: Cart Feature	R,A	I	I	C	I
Feature 14: Question Bank Feature	A	C	R	C	I
Feature 15: Quiz Feature	A	R	I	R, C	I

Feature 16: Section Feature	A	I	I	R, C	R
Feature 17: Lecturer Application Feature	R	R	I	C	I
Feature 18: Question Feature	A	R	R	R, C	I
Feature 19: Answer Feature	A	R	R	R, C	I
Feature 20: Rating Feature	A	I	I	R, C	R
Feature 21: Student Progress Feature	A	R	I	R, C	I
Feature 22: Payment Feature	R,A	I	I	C	I
Feature 23: Transaction Feature	R,A	I	R	C	I
Feature 24: Banner Feature	A	I	R	R, C	I
Feature 25: User Feature	A	R	I	C	I
Feature 26: Lecturer Feature	A	R	I	C	R
Feature 27: Admin Feature	A	R	I	C	R

Work Package	H E1 76 38 2	H E 1 7 3 2 4 4	H E1 8 6 6 2 1	H E 1 7 2 2 3 8	H E 1 6 3 3 4 4
Report 1: Project Introduction	R, A	R	R	R,C	R
Report 2: Project Management Plan	R, A	R	R	R,C	R
Report 3: Software Requirement Specification	R, A	R	R	R,C	R
Report 4: Software Design Specification	R, A	R	R	I,C	R
Report 5: Test Documentation	R, A	R	R	R,C	R
Report 5.1: Unit Test	R, A	R	R	I,C	R
Report 5.2: Integration Test	I, A	I	I	R,C	I
Report 5.3: System Test	I, A	I	I	R,C	I
Report 6: Software User Guides	I, A	I	I	R,C	I
Report 7: Final Project Report	R, A	R	R	R,C	R
Meeting & review	R, A	R	R	R,C	R
Project Management Overhead	R, A	R	R	R,C	R

4. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Daily Meeting	All team members	Report tasks that have been done. Discuss issues, resolve problems. Define the next task	21:00 every week	Online (Google Meet)
Quick Messaging - Group chat	All team members	When you need to have a quick casual talk or schedule a meeting with a supervisor or among team members.	Every day	Online (Zalo)
Weekly meeting with the supervisor	Team Members, Supervisor	Report the project's progress, log issue, define next week plan	21:00 on Sunday, every week	Offline, Online (Google Meet)

5. Configuration Management

5.1 Document Management

Regarding documents, we will store documents related to this project as well as documents related to meetings in Google Drive, Zalo. At the same time, some important documents will also have a copy on GitHub.

5.2 Source Code Management

Regarding source code management, we chose GitHub as the place to store project versions because GitHub is a popular and easy-to-use tool.

5.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	ReactJS(FrontEnd), .NET(BackEnd), Python(Backend)
Database	SQL Server, Qdrant
IDEs/Editors	Visual Studio Code (FrontEnd), Visual Studio 2022 (BackEnd)
Diagramming	DrawIO, Visual Paradigm, Lucidchart
Documentation	Google Docs/Sheets/Canva/Powerpoint
Version Control	GitLocal/Github (Source Codes), Google Drive (Documents)
Deployment server	Vercel (Frontend), Vietnix Server (Backend)
Project management	Github, Zalo, Google Drive